

[By Sanjay]

- **Making Steps of Project**

1. Collect India climate data (multiple sources) .
2. Make a folder with the name of climate-digital-twin .
3. Make another folder for Store row data and .
4. Do some analysis and check if the data is suitable or not for the model .
5. Perform Pandas and numpy and seaborn and matplotlib for analysis .
6. Install all dependency and do analysis After analysis .
7. Make data for Machine learning ready.
8. Apple feature engineering and do column operation like Standardization or One hot encoding etc.
9. Then use different models for train model Gradient boating or random forest.
10. Then do prediction on both models and after seeing the result on graph or numeric .
11. Decide to go with gradient boating model
12. And also train Model store both models in the model folder.
13. Also ready reports or Dashboard For better understanding

- **[Step i do for Train ML model (INFomation about Model)]**

CLIMATE DIGITAL TWIN - INDIA

Analysis Report

Date: 2026-06-17 13:47:09

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DATASET STATISTICS

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Total Records: 7310

Cities Covered: 5

Date Range: 2024-01-01 to 2044-01-05

Duration: 7309 days

Temperature:

- Average: 26.6°C

- Range: 18.0°C to 35.0°C

Humidity:

- Average: 60.1%

- Range: 30.0% to 90.0%

Rainfall:

- Total: 36206.9 mm

- Daily Average: 5.0 mm

Air Quality Index:

- Average AQI: 175.4
- Poor Days: 1813
- Moderate Days: 1846
- Good Days: 1871

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FEATURE ENGINEERING

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Total Features Created: 22

Feature Categories:

- Temporal Features: 7
- Lag Features: 4
- Rolling Features: 2
- Derived Features: 4
- Categorical: 1

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EXTREME EVENTS

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Heat Waves: 198 days
Cold Waves: 731 days
Heavy Rainfall: 731 days
High AQI: 2936 days
Strong Winds: 731 days

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ML MODEL PERFORMANCE

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Training Samples: 5844
Testing Samples: 1461

Random Forest:

- RMSE: 5.030°C
- MAE: 4.368°C
- R² Score: -0.0040

Gradient Boosting (BEST):

- RMSE: 5.125°C
- MAE: 4.414°C
- R² Score: -0.0423

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RECOMMENDATIONS

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1. Use Gradient Boosting model for production (best R^2 score)
2. Implement LSTM for long-term forecasts
3. Deploy Kalman Filter for continuous data assimilation
4. Create real-time dashboard with alerts
5. Set up monthly retraining pipeline

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